熊宇琦

电话：(+86) 137-9850-7966 | **邮箱：**yqbear426@gmail.com | **微信：**xyq426_ | **主页：**YukiBear426.github.io

教育背景

深圳大学 | 电子与信息工程学院 2022.09 - 2026.06

- GPA：4.09 / 4.5，平均分：**90.0 / 100**，排名：**6 / 246**，六级：574，托福：95
- 核心课程：大数据分析 (99)、数字信号处理 (97)、智能语音处理 (96)、电路分析 (96)、人工智能 (96)、面向对象编程 (95)、Python 编程 (94)、C 语言 (93)、模电/数电 (93)、图神经网络 (93)、深度强化学习 (92)、迁移学习 (92)、机器视觉技术 (92)

新加坡国立大学 | 电气与计算机工程学院 2026.07 - 至今

- 核心课程：机器学习、视觉计算、模式识别、天线工程、电磁波理论
- 研究项目：研究 AI 生成图像检测的可靠性与跨生成工具泛化能力，分析检测器对图像伪影及后处理操作的依赖。构建评测流程，开展基准测试与失败案例分析，探索提升检测鲁棒性的策略。

实习经历

中国电子科技集团智慧城市研究院 2024.11 - 2025.05

- 优化表格识别模型：
  - 优化基于 YOLO、OpenCV 与 OCR 的表格识别模型，完成表格检测、区域裁剪、结构解析、单元格提取及文本识别，实现图像表格向可编辑 Excel 的自动转换，有效降低人工数据录入成本。
- 开发基于大语言模型的金融数据问答系统：
  - 对历史财务数据进行清洗与标准化，导入关系型数据库 MySQL；利用规则结合 BERT 预训练模型实现用户问句的意图识别与关键实体抽取，并将解析结果映射为 SQL 模板执行查询，将结果转化为自然语言返回，实现金融数据问答。
- 构建智能化安全态势感知系统：
  - 参与构建基于大语言模型（LLMs）的智能化安全态势感知系统，针对跨域异构日志进行深度语义分析与意图识别，通过设计专家级 Prompt 工程与 Few-shot 策略，成功解决复杂场景下的安全识别及实时响应难题。

科研经历

非平稳时间序列预测 2024.09 - 2025.02

- 项目概述：
  - 针对非平稳时间序列预测中现有模型无法有效捕捉随时间变化的统计特性、预测精度下降的问题，本项目提出 AEFIN 框架，通过傅里叶变换将时间序列分解为平稳和非平稳部分，并引入交叉注意力机制增强两部分间信息共享；结合时域与频域特征设计自适应损失函数，提升非平稳序列预测精度和模型鲁棒性。
- 成果产出：
  - 以第一作者身份，在 2025 年国际神经网络联合会议 (IJCNN) 发表论文《Non-Stationary Time Series Forecasting Based on Fourier Analysis and Cross Attention Mechanism》，论文链接：https://ieeexplore.ieee.org/document/11228920

显著性目标检测 2025.02 - 2025.05

- 项目概述：
  - 针对显著性目标检测在复杂场景下细节丢失、边缘模糊及单模态信息融合不足的问题，本项目通过动态不确定性图卷积模块增强小结构和边缘区域检测精度，并结合多模态协同融合策略对 RGB、深度和边缘特征加权融合，实现跨模态语义互补与一致性。该方法在标准数据集上显著提升边缘清晰度和复杂背景下的鲁棒性。
- 成果产出：
  - 以第一作者身份，在 2025 年第 32 届国际神经信息处理会议 (ICONIP) 发表论文《Graph-Based Uncertainty Modeling and Multimodal Fusion for Salient Object Detection》，论文链接：https://link.springer.com/chapter/10.1007/978-981-95-4445-5_9

多无人机系统任务分配与路径规划 2025.09 - 2026.05

- 项目概述：
  - 针对多无人机系统在能量受限及动态环境下的任务执行挑战，本项目设计了一套基于能耗代价函数的路径规划与多任务分配方法，深入分析任务分配与路径规划之间的耦合效应。引入协同效率比（SER）对算法组合间的非加性交互机制进行量化，为多无人机协同系统的设计提供理论支撑。

获奖经历

- 全国大学生数学竞赛，三等奖（2024）
- 粤港澳大湾区数据应用与创新竞赛，三等奖（2024）
- 深圳大学学习之星一等 (前 **1%**) 奖学金（2024）
- 深圳大学学习之星二等 (前 **3%**) 奖学金（2023 & 2025）
- 深圳大学双创之星二等 (前 **3%**) 奖学金（2025）
- 深圳大学校级优秀毕业生 & 荣誉学士学位（2026）

Yuqi Xiong

(+86)137-9850-7966 yqbear426@gmail.com xyq426_ github.com/YukiBear426

EDUCATION

ShenZhen University **Shenzhen, China**
Bachelor of Electronics and Information Engineering *2022.09 - 2026.06*

- GPA: 4.09/4.5, Average Score: **90.0/100**, Ranking: **6/246**, CET-6: **574**, TOEFL: **95**
- Core Courses: Big Data Analysis(99), Digital Signal Processing(97), Intelligent Speech Processing(96), Circuit Analysis(96), Artificial Intelligence(96), Object-Oriented Programming(95), Python Programming(94)

National University of Singapore **Singapore**
Master of Science in Electrical and Computer Engineering *2026.08 - Present*

- Core Courses: Machine Learning, Visual Computing, Pattern Recognition, Antenna Engineering, EM Wave Theory

INTERNSHIP EXPERIENCE

Smart City Research Institute of China Electronics Technology Corporation **Shenzhen, China**
Research Assistant *2024.11 - 2025.05*

- Optimized table recognition model using YOLO, OpenCV, and OCR; performed table detection, region cropping, structural parsing, cell extraction, and text recognition, converting image-based tables into editable Excel files and reducing manual data entry.
- Developed a financial data QA system by cleaning and standardizing historical financial data and importing it into a MySQL relational database. Implemented a rule- and BERT-based approach for user query intent classification and key entity extraction, mapping parsed queries to SQL templates for database retrieval and converting results into natural language responses.
- Engineered an LLM-based intelligent security situational awareness system by performing deep semantic analysis and intent recognition on cross-domain heterogeneous logs. Designed expert-level prompt engineering and few-shot learning strategies to address security identification and real-time response challenges in complex scenarios.

RESEARCH EXPERIENCE

Non-stationary Time Series Forecasting **Supervisor: Hao Wang**
Research Assistant *2024.09 - 2025.02*

- Proposed AEFIN, a novel framework that combines Fourier Transform to decompose time series into stationary and non-stationary components, cross-attention mechanisms to enhance information sharing, and Fourier analysis to capture periodic patterns, significantly enhancing the prediction accuracy.
- Designed a custom loss function that uses MSE for stable time-domain and frequency-domain components, MAE for unstable time-domain components, which are weighted and integrated to enhance model robustness.
- Achieved SOTA results on most datasets, outperforming common models in both MSE and MAE metrics when combined with existing mainstream transformer-based or CNN-based time series forecasting models.

♡ **Yuqi Xiong**, et al. "Non-Stationary Time Series Forecasting Based on Fourier Analysis and Cross Attention Mechanism." *2025 International Joint Conference on Neural Networks (IJCNN), Rome, Italy.*

Salient Object Detection **Supervisor: Yang Wen**
Research Assistant *2025.02 - 2025.05*

- Developed a dynamic uncertainty-aware graph convolution module to enhance the detection of fine structures and edge regions in salient object detection, addressing detail loss and blurred boundaries in complex scenes.
- Designed a multimodal fusion strategy to adaptively integrate RGB, depth, and edge features, enabling cross-modal semantic complementarity and consistency.
- Experiments on standard datasets demonstrate improved edge clarity and robustness against complex backgrounds.

♡ **Yuqi Xiong**, et al. "Graph-Based Uncertainty Modeling and Multimodal Fusion for Salient Object Detection." *The 32nd International Conference on Neural Information Processing (ICONIP), Okinawa, Japan.*

Multi-UAV Task Allocation& Path Planning **Supervisor: Bharadwaj Veeravalli**

Research Assistant *2025.09 - 2026.05*

- To address the challenges of task execution in multi-UAV systems under energy constraints and dynamic environments, this project proposes a path planning and task allocation method based on an energy cost function.
- It provides an in-depth analysis of the coupling effects between task allocation and path planning.
- The Synergistic Efficiency Ratio (SER) is introduced to quantify the non-additive interaction mechanisms among algorithm combinations, offering theoretical support for the design of cooperative multi-UAV systems.

♡ **Yuqi Xiong**, et al. "Synergy Analysis of Task Allocation and Path Planning Algorithms for Energy-Aware Multi-UAV Systems." (Manuscript in submission).

HONORS & AWARDS

Chinese Mathematics Competition, Third Prize (2024)
Greater Bay Area Data Application and Innovation Challenge, Third Prize (2024)
Shenzhen University First-Class **(Top 1%)** Academic Scholarship (2024)
Shenzhen University Second-Class **(Top 3%)** Academic Scholarship (2023 & 2025)
Shenzhen University Second-Class **(Top 3%)** Innovation and Entrepreneurship Scholarship (2025)
Shenzhen University Outstanding Graduate & Bachelor's Degree with Honours (2026)

Skills

Programming Languages: Proficient in Python; familiar with C and MATLAB
Technical Skills: Machine Learning, Deep Learning, Data Mining, Computer Vision, Image Processing
Tools & Platforms: Proficient in PyTorch, basic knowledge of Linux, Git, TensorFlow